NATURAL CAUSES OF CLIMATE VARIABILITY

Natural variability refers to climate variations caused by processes other than human influence. It includes both variations generated within the climate system and variations forced by natural factors outside the system. The IPCC emphasises that natural variability is a major source of year-to-year climate change and may remain prominent across several years or even decades.

Natural variability is not the same as a persistent long-term climate trend. A short period may show rapid warming, weak warming or temporary cooling because natural fluctuations are superimposed on the longer-term trend. When the averaging period becomes long enough, these fluctuations increasingly cancel or average out, allowing the underlying forced trend to become clearer.

Two dimensions of natural variability

- Origin: whether the process is generated internally within the atmosphere-ocean-ice-land system or driven externally by natural changes.
- Timescale: whether its effects are expressed over years, decades, multiple decades or thousands of years.

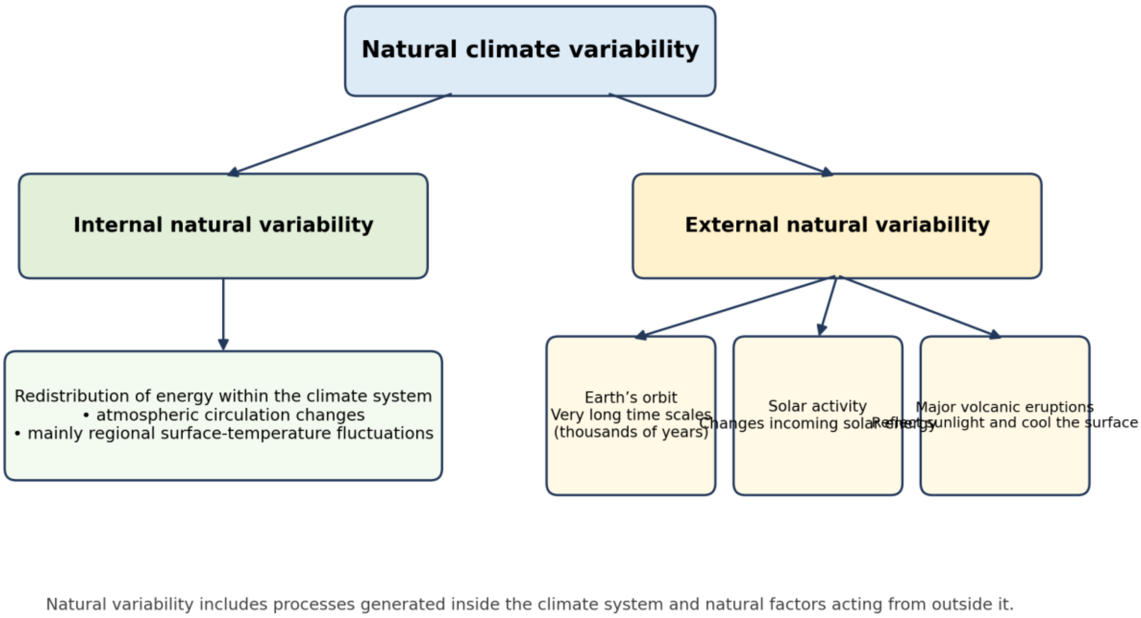


Figure 1. Classification of natural climate variability into internal and external components.

Categories of Natural Climate Variability

1. Internal natural variability

Internal natural variability is generated within the climate system. It represents a redistribution of energy rather than an external addition or removal of energy. The document gives atmospheric circulation changes, similar to processes that produce daily weather, as an example. It also refers to El Nino and similar variations when discussing natural-only climate simulations.

- It is produced by interactions among components of the climate system.

- It is especially visible in regional surface-temperature fluctuations.
- It contributes strongly to annual variability and can affect trends over several years or a couple of decades.
- Because it redistributes energy, its global long-term effect is smaller than its regional and short-term effect.

2. External natural variability

External natural variability is driven by naturally occurring changes outside the internally evolving climate system. The three external natural factors: changes in Earth's orbit, small variations in energy received from the Sun, and major volcanic eruptions.

Table 1. Natural causes

Cause	Category	Mechanism described	Characteristic timescale	Importance for recent warming
Internal climate variability	Internal	Redistributes energy within the climate system through circulation and related processes.	Years to decades; strongest regionally and over short intervals.	Can amplify or weaken short-term warming but has only a mild effect on long multi-decadal trends.
Changes in Earth's orbit	External	Alter the configuration of Earth relative to incoming solar energy.	Thousands of years.	Very little change during the past century; very little influence on recent temperature change.
Variations in solar activity	External	Alter the amount of incoming energy received from the Sun.	Discussed as a natural driver over decades to centuries.	Included in natural-only simulations, but cannot reproduce the observed strong warming by itself.
Major volcanic eruptions	External	Add aerosols to the upper atmosphere; these reflect sunlight and cool the surface.	Strong but short-lived; effect generally fades within a decade.	Can produce temporary cooling, not the sustained historical warming trend.

Natural Causes and Their Climatic Effects

1. Changes in Earth's orbit

Large changes in Earth's orbit are associated with major global climate changes in the distant past. Their defining characteristic is their very long operating timescale - thousands of years. Because orbital conditions changed very little during the past century, the orbital change had very little influence on the temperature change observed over that period.

2. Variations in solar activity

Variations in solar activity change the quantity of incoming energy received by Earth. The solar variability as one of the two main natural drivers of climate variation over decades to centuries, along with major volcanic eruptions. Solar changes are included in climate-model experiments that represent natural forcings only; however, these simulations do not reproduce the strong observed warming.

3. Major volcanic eruptions

Large volcanic eruptions inject small particles, or aerosols, into the upper atmosphere. These particles increase the reflection of sunlight and therefore cool Earth's surface. The cooling can be strong, but it is temporary. The influence on surface temperature generally fades within a decade of the eruption.

4. Internal circulation and energy redistribution

Internal climate variability operates through changes in atmospheric and related circulation processes. It shifts energy among regions and components of the climate system. This is why the clearest surface-temperature effects are frequently regional rather than global, and why short-term global temperature can fluctuate around the long-term forced trend.

Quantitative Contribution of Natural Variability Since 1850

For the historical period beginning around 1850, the natural variability contributed between -0.23°C and +0.23°C to global surface-temperature change. During approximately the same period, observed warming was about 1.1°C.

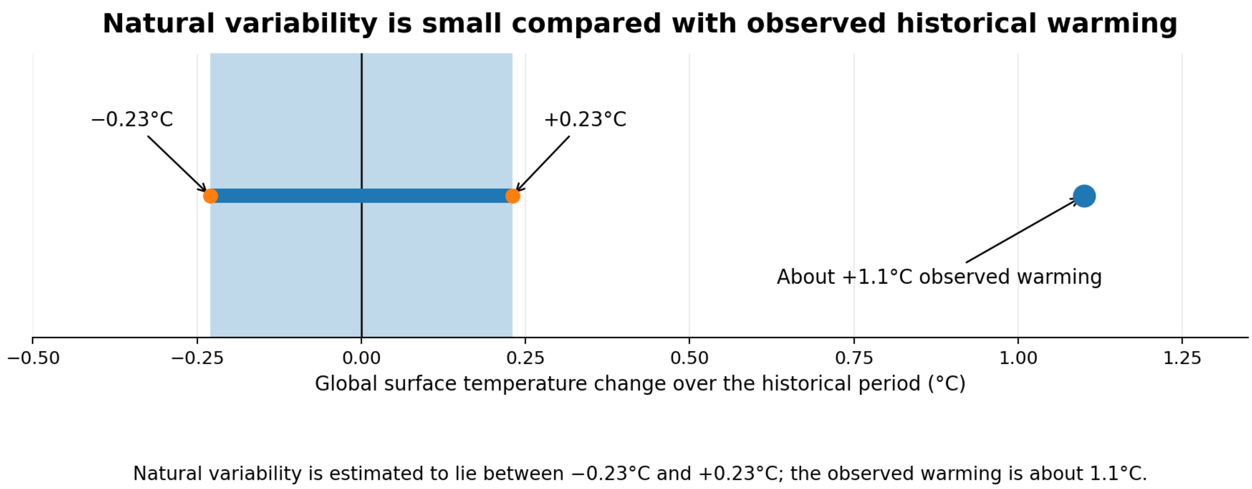
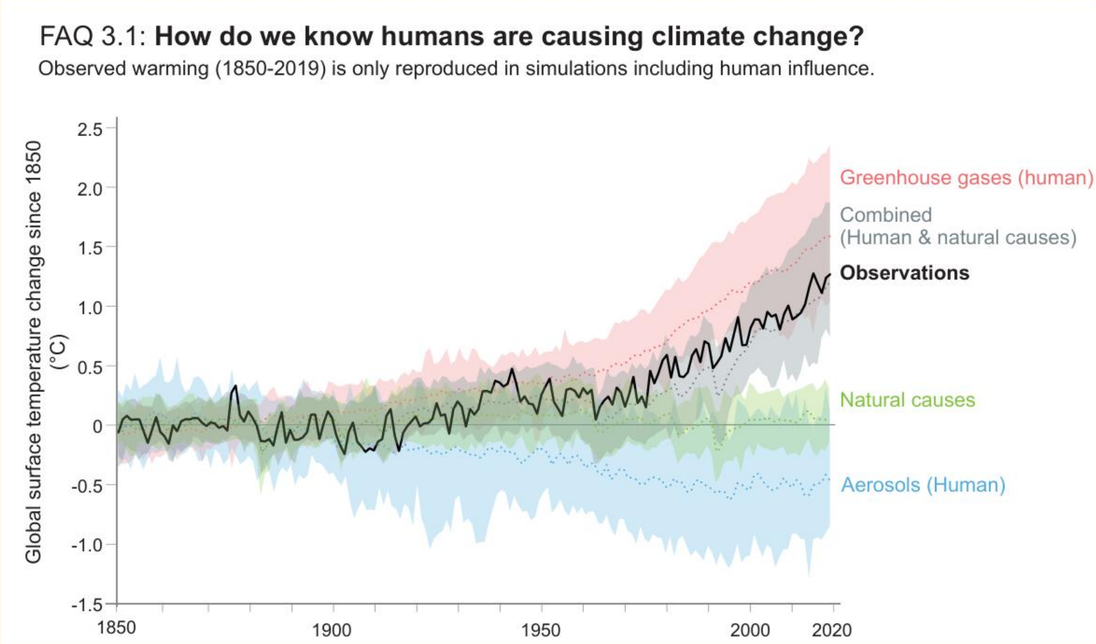


Figure 2. Estimated contribution of natural variability compared with observed historical warming.

Interpretation of the range

1. The negative end (-0.23°C) represents a possible net cooling contribution from natural variability over the historical period.
2. The positive end (+0.23°C) represents a possible net warming contribution.
3. Both values are much smaller than the observed warming of about 1.1°C.
4. Therefore, natural variability can alter the exact magnitude and timing of warming, but it cannot account for most of the historical increase.

FAQ 3.1 (continued)



FAQ 3.1, Figure 1 | Observed warming (1850–2019) is only reproduced in simulations including human influence. Global surface temperature changes in observations, compared to climate model simulations of the response to all human and natural forcings (grey band), greenhouse gases only (red band), aerosols and other human drivers only (blue band) and natural forcings only (green band). Solid coloured lines show the multi-model mean, and coloured bands show the 5–95% range of individual simulations.

Figure 3. Observed warming is reproduced only when human influence is included in the simulations.

HUMAN ACTIVITIES CAUSING CLIMATE CHANGE

Climate change and human influence
Climate change refers to long-term shifts in temperature and weather patterns. Day-to-day weather varies naturally, but climate concerns persistent changes in averages, seasons, extremes and related environmental conditions over longer periods.

Recent studies acknowledge that natural processes influence climate. However, they consistently emphasise that the rapid change associated with the industrial era is primarily linked to human activities that release carbon dioxide and other heat-trapping gases. These emissions alter atmospheric composition, retain additional energy in the Earth system and produce global warming.

Central proposition

Human activities do not create the greenhouse effect from nothing. They strengthen a natural heat-retaining process by increasing the concentration of greenhouse gases and by changing the reflectivity and energy behaviour of the land surface.

- Industrial-era change: Large-scale fossil-fuel combustion and industrial development increased greenhouse-gas emissions.
- Atmospheric change: The added gases trap a greater portion of outgoing infrared radiation.
- System response: Heat accumulates in the atmosphere, land and oceans; ice melts and sea level rises.
- Extreme-event context: Natural weather systems still create individual events, but warming can push them beyond earlier ranges and design thresholds.

Principal anthropogenic pathways

The two broad pathways through which human activities affect climate:

- Changing atmospheric composition through emissions of greenhouse gases and other climate-active substances.
- Changing land-surface properties, particularly reflectivity (albedo), heat absorption, vegetation cover and the movement of water and energy at the surface.

Human activity	Primary climate linkage	Representative result
Burning coal, oil and natural gas	Emits CO_2 and supports energy, transport and industrial activity.	Higher atmospheric CO_2 and stronger infrared heat retention.
Industrialisation	Uses fossil energy and may release CO_2 and fluorinated gases.	Large contribution from fossil-fuel and industrial-process CO_2.
Urbanisation and construction	Replaces vegetation and permeable land with buildings, roofs, pavement, bitumen and concrete.	Lower reflectivity or greater heat absorption; urban heat-island effect.
Deforestation	Removes vegetation and changes surface reflectivity and land carbon storage.	CO_2 emissions from forestry and other land-use change; altered local energy balance.
Agriculture	Changes land cover; rice cultivation emits methane; fertilised systems are associated with nitrous oxide in the sectoral framing.	CH_4 and N_2O increases, plus regional albedo and moisture changes.
Animal husbandry	Livestock activities as a methane source.	Rising atmospheric methane.
Transport	Uses petroleum fuels and contributes to sectoral greenhouse-gas emissions.	CO_2 emissions and urban air/heat burdens.
Refrigeration and specialised industrial gases	HFCs, PFCs and SF_6 among the major controlled greenhouse gases.	Very strong and, in some cases, long-lived heat-trapping effects.

Sector-wise human causes of climate change

The climate change is not caused by a single activity. Energy supply, land use, industry, transportation and buildings form an interconnected emissions system.

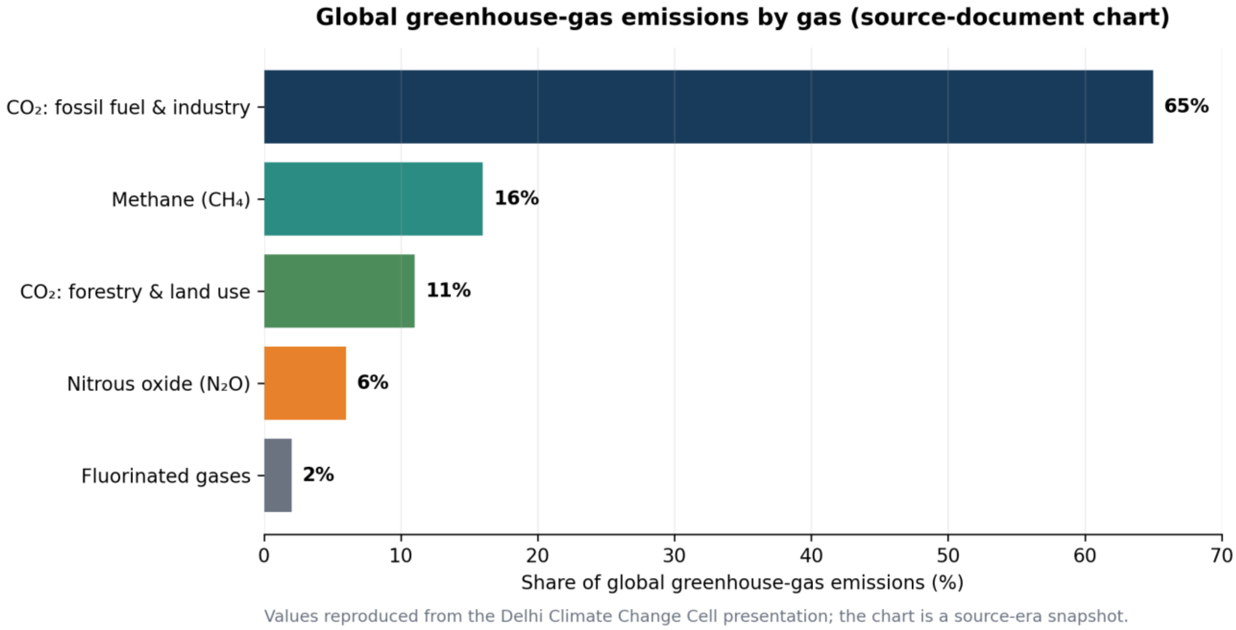


Figure 3. Global greenhouse-gas emissions by gas

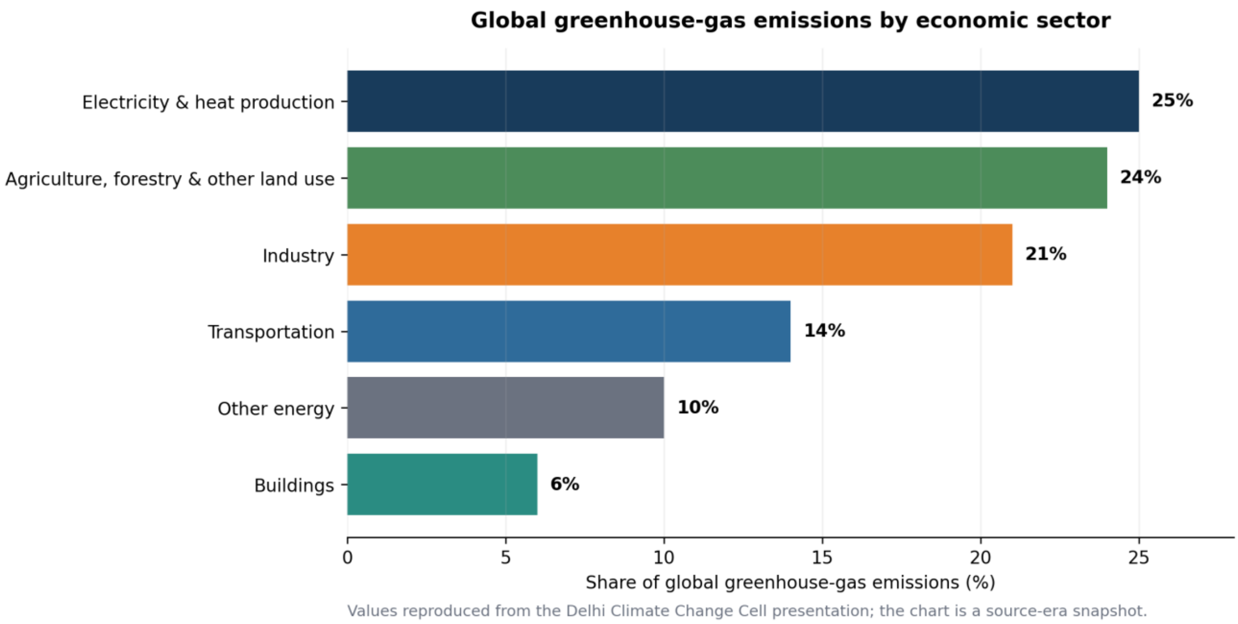


Figure 4. Global greenhouse-gas emissions by economic sector

Sector	Share	Why it matters in the causal chain
Electricity and heat production	25%	Fossil fuels burned for power and heat release large quantities of CO_2.
Agriculture, forestry and other land use	24%	Combines methane and nitrous oxide emissions with deforestation and land-carbon changes.
Industry	21%	Uses energy and contributes process-related emissions; fluorinated gases are also linked to industrial applications.
Transportation	14%	Petroleum-fuel combustion produces CO_2 and supports urban emissions growth.
Other energy	10%	Includes additional energy-system activities beyond electricity and heat production.
Buildings	6%	Operational energy use and urban materials contribute to emissions and local heat storage.

Atmospheric evidence of human influence

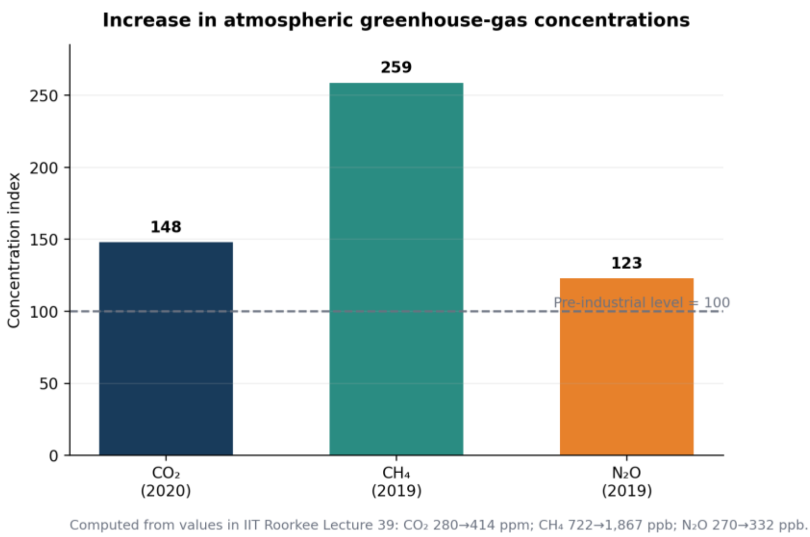


Figure 5. Relative increase in CO_2, CH_4 and N_2O concentrations

Gas	Pre-industrial / eighteenth-century value	Later value	Approximate change
CO_2	280 ppm	414 ppm (2020)	About 48% higher.
CH_4	722 ppb	1,867 ppb (2019)	About 159% higher.
N_2O	270 ppb	332 ppb (2019)	About 23% higher.

How human-caused warming alters extremes and impacts
The connect additional heat in the Earth system to changes in extremes, oceans, ice, ecosystems and society.

Heat waves

Heat waves usually arise from natural weather patterns such as strong, slow-moving anticyclones. Human-caused warming raises the background temperature, making an already hot event more likely to exceed historical records and physiological or engineering thresholds. The source article projects more frequent, more intense and longer-lasting heat waves under continued warming.

Drought and wildfire

During drought, persistent high pressure suppresses rainfall. Added warming dries soil and vegetation, reduces evaporative cooling and raises atmospheric moisture demand. This reinforces heat and increases wildfire risk. The result is a positive feedback in which dryness and heat intensify one another.

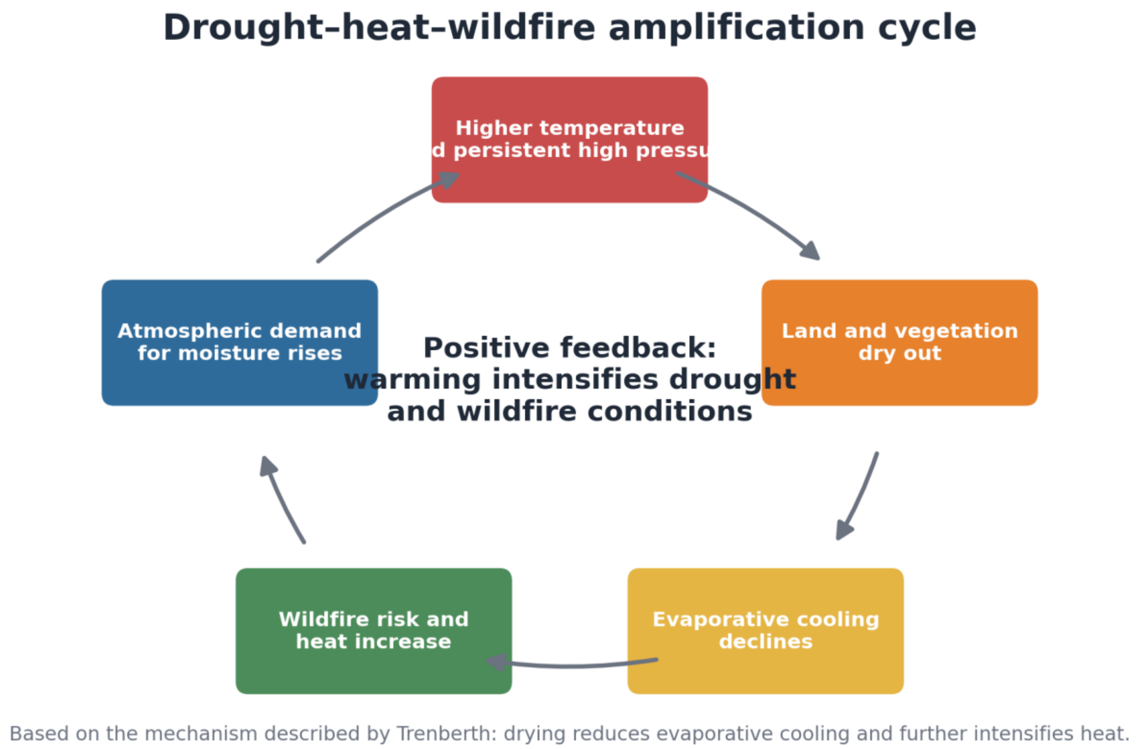


Figure 8. Drought-heat-wildfire amplification cycle derived from the mechanism described by Trenberth.

Heavy rainfall, flooding and storms

The atmosphere can hold about 7% more moisture per 1°C of warming. Warmer oceans supply additional evaporation and water vapour. Storms can therefore access more moisture, increasing the potential for intense rainfall, flooding and latent-heat release. The precipitation-intensity increases of approximately 5-20% in relevant events, while stressing that impacts vary among storms.

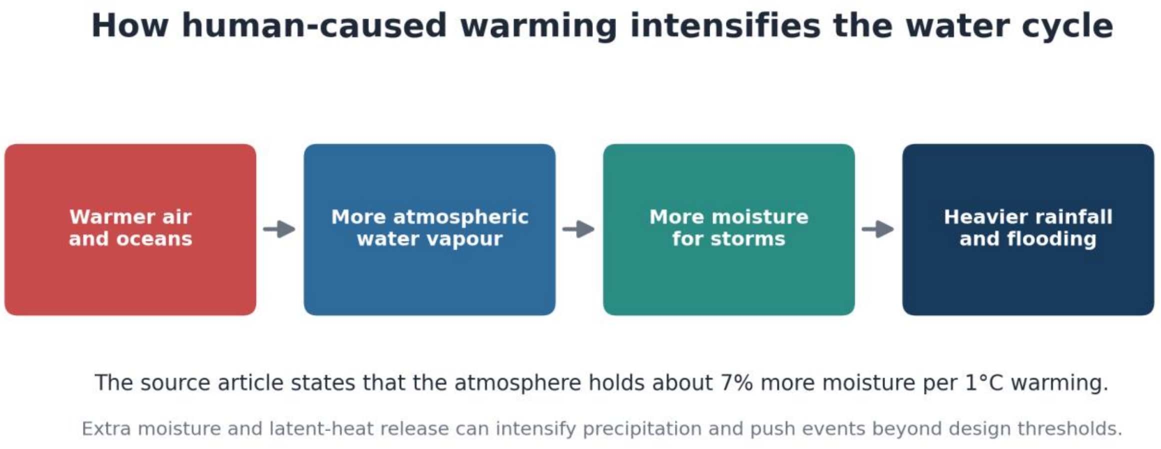


Figure 9. Mechanism linking warming, atmospheric moisture and heavier precipitation.

Tropical storms, sea level and coastal risk

Tropical storms draw energy from warm oceans and the release of latent heat in heavy rainfall. The source article explains that warming can contribute to heavier rain, greater intensity and more damaging storm surge when combined with higher sea level. Large natural variability makes trend detection difficult, but the physical warming and moisture mechanisms remain important.

Snow, ice and seasonal water

The sources describe melting glaciers and sea ice, earlier snowmelt and a shorter snow season at its margins. Warmer conditions can cause more precipitation to fall as rain. Earlier melt changes runoff timing and may increase summer water shortage and wildfire risk. At the same time, a warmer, moister atmosphere can still produce heavy mid-winter snowfall when temperatures remain below freezing.

Human-caused change	Physical mechanism	Examples of outcomes described
Higher background temperature	Additional retained infrared energy.	Heat waves, warmer land and oceans, stress on health and infrastructure.
More atmospheric moisture	Warmer air holds more water vapour; warm oceans increase evaporation.	Heavier rainfall, flooding and stronger precipitation in storms.
Drier land under persistent heat	Evaporation increases; plants wilt and evaporative cooling declines.	Drought intensification, wildfire and forest stress.
Warmer oceans and higher sea level	Greater ocean heat content and thermal expansion; ice melt.	More damaging storm surge and coastal flooding.
Changing snow and ice state	Small temperature changes shift water between solid and liquid forms.	Earlier snowmelt, glacier retreat and altered seasonal water supply.

Source basis: Trenberth (2018), pp. 1-6; IIT Roorkee Lecture 39, pp. 3 and 10-19; Delhi Climate Change Cell, pp. 10-39.

Why modest climate shifts can cause large damage
Trenberth emphasises that the climate contribution to an extreme event may appear modest in percentage terms, yet damage can increase nonlinearly once a threshold is crossed.

Infrastructure, drainage systems, electricity networks, crops, ecosystems and the human body are often designed or adapted to a historical range. A small shift can move an event beyond that range.

Threshold example	Small climatic shift	Possible nonlinear consequence
Urban drainage	A short-duration rainfall event becomes somewhat more intense.	Drainage capacity is exceeded, causing widespread street and property flooding.
Heat tolerance	Background temperature rises during a natural heat wave.	Health emergencies and electricity demand rise sharply; transport and materials may fail.
Wildfire conditions	Vegetation becomes slightly drier and hotter.	Fire ignition, spread and suppression costs increase disproportionately.
Coastal protection	Sea level is higher when a storm surge arrives.	Water overtops barriers that previously contained comparable storms.
Agriculture	A crop is exposed to heat or drought beyond a physiological threshold.	Yield loss can be much larger than the percentage change in average climate.

Industrialization as a Cause of Climate Change
Industrialization is the transition from small-scale, labour-based production to large-scale, machine-based manufacturing supported by factories, electricity, transport networks, mining and mass consumption.

Industrialization has improved production, infrastructure, employment and living standards. However, it has also greatly increased the extraction and combustion of coal, oil and natural gas. These activities release greenhouse gases and modify land surfaces, thereby affecting Earth's energy balance.

Industrialization and the Industrial Revolution
Before industrialization, societies depended mainly on human labour, animal power, biomass, and small-scale water and wind energy. The Industrial Revolution transformed this system by introducing mechanised production, coal-powered steam engines, large factories, extensive mining, and large-scale iron and steel production. It also led to the rapid expansion of railway and road transport, mass production of cement, chemicals and consumer goods, and large-scale electricity generation.

These developments sharply increased energy and material consumption. The atmospheric greenhouse-gas concentrations remained comparatively stable for centuries but began increasing rapidly after the industrial era because of fossil-fuel burning.

The IPCC concludes that human activities, primarily greenhouse-gas emissions, have unequivocally caused global warming. The global surface temperature in 2011–2020 was approximately 1.1°C above the 1850–1900 level.

Industrialization is a system of interconnected activities

Industrialization should not be understood only as emissions from factory chimneys. It creates an interconnected system involving:

Industrial development
↓
Greater demand for energy and raw materials
↓
Coal, oil and natural-gas extraction
↓
Electricity generation and industrial combustion
↓
Production of steel, cement, chemicals and goods
↓
Expansion of transport, buildings and cities
↓
Higher greenhouse-gas emissions
↓
Enhanced greenhouse effect
↓
Climate change

Therefore, industrialization influences climate through both direct and indirect pathways.

Principal pathways linking industrialization to climate change

Fossil-fuel combustion

Factories require substantial energy to operate machinery, heat furnaces and boilers, generate steam and electricity, and carry out the drying, melting and processing of materials. Energy is also required for pumping, cooling, ventilation, and the transportation of raw materials and manufactured products.

Coal, petroleum products and natural gas contain carbon. During complete combustion, carbon combines with oxygen and forms carbon dioxide:

$$C + O_2 \rightarrow CO_2 + \{energy\}$$

For methane or natural gas:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{energy}$$

The energy is useful for production, but the CO_2 enters the atmosphere. The fossil-fuel combustion for meeting energy demand as a major cause of greenhouse-gas emissions.

Electricity and heat consumption

Industrial facilities consume large quantities of electricity. When electricity is produced from coal, oil or natural gas, emissions occur at the power plant rather than inside the factory. These are called indirect industrial emissions.

For example:

Factory demands electricity
↓
Coal-based power plant generates electricity
↓
Coal combustion releases CO_2
↓
Electricity is supplied to the factory

Although the emissions occur at the power station, industrial demand is partly responsible for them.

Industrial-process emissions

Some industrial emissions arise from the chemical transformation of raw materials. These emissions occur even when the process heat is supplied by renewable electricity.

Important examples of industries that generate process-related greenhouse-gas emissions include cement manufacturing, iron and steel production, lime production, chemical and petrochemical manufacturing, and the production of aluminium and other metals.

The IPCC classifies industrial emissions as arising from fuel combustion, industrial processes, product use and waste. Together, these represented approximately 24% of direct anthropogenic greenhouse-gas emissions in 2019.

Cement production

Cement is essential for concrete, buildings, roads, bridges and other infrastructure. Industrialization and urban expansion therefore increase cement demand.

Cement production causes CO_2 emissions through two mechanisms.

Fuel-related emissions

Kilns must operate at very high temperatures. Fossil fuels are commonly burned to provide this heat.

Fossil fuel combustion in kiln
↓
Heat for clinker production
↓
CO_2 from fuel combustion

Calcination or process emissions

Limestone consists mainly of calcium carbonate. During clinker production, limestone is heated and decomposes:

$$CaCO_3 \rightarrow CaO + CO_2$$

where:

- $CaCO_3$ = calcium carbonate, commonly known as limestone
- CaO = calcium oxide, commonly known as lime
- CO_2 = carbon dioxide released during the reaction

This CO_2 is produced by the chemical reaction itself. It cannot be eliminated simply by replacing coal with renewable electricity.

The IPCC estimates that approximately 60% of direct cement-production emissions arise from limestone decomposition, while around 40% arise from process heating.

Iron and steel production
Steel is an essential material used in the construction of buildings, bridges and railways. It is also widely required for manufacturing machines, automobiles, industrial equipment, pipelines and power infrastructure.

Conventional primary steel production commonly uses coke, which is produced from coal. Coke acts both as a fuel and as a reducing agent that removes oxygen from iron ore.

A simplified reduction reaction is:

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

where:

- Fe_2O_3 = iron oxide
- CO = carbon monoxide
- Fe = iron
- CO_2 = carbon dioxide

The dominant blast furnace–basic oxygen furnace route is more emissions-intensive than producing steel from recycled scrap using electric arc furnaces. The IPCC identifies steel as one of the most important industrial sources of greenhouse-gas emissions.

Chemical and petrochemical industries
Industrialization increases the production of plastics, fertilisers, ammonia, solvents, synthetic fibres, paints, detergents, pharmaceuticals and various industrial chemicals.

These industries contribute to climate change because they:

1. consume large quantities of electricity and heat;
2. use oil, natural gas or coal as fuels;
3. use fossil carbon as a raw material or feedstock;
4. release greenhouse gases during chemical reactions;
5. produce carbon-intensive materials and products.

Some fossil carbon remains stored temporarily in products such as plastics. It may later enter the atmosphere when products are burned or decompose after disposal.

Fluorinated industrial gases
Industrial activities also produce or use fluorinated greenhouse gases, including hydrofluorocarbons (HFCs), perfluorocarbons (PFCs) and sulphur hexafluoride (SF_6).

They may be released from industrial equipment, production processes and product use. Although their atmospheric concentrations are much lower than that of CO_2, many fluorinated gases have strong heat-trapping effects and may remain in the atmosphere for long periods.

Mining and raw-material extraction
Industrial production requires large quantities of raw materials and energy resources, including coal, petroleum, natural gas, iron ore, limestone, bauxite, sand, aggregates and various other minerals.

Mining and extraction contribute to climate change through the fuel used in excavation and drilling, electricity consumed by machinery, and energy required for crushing, grinding and processing materials. Additional emissions arise from transporting raw materials and from methane released during coal mining and other fossil-fuel operations. These activities may also remove vegetation and disturb land, reducing carbon storage and altering local environmental conditions.

Thus, emissions begin before the raw material reaches the manufacturing plant.

Transportation created by industrialization
Industrialization requires transportation at every stage:

Raw-material extraction
↓
Transport to processing plant
↓
Transport to manufacturing facility
↓
Distribution to markets
↓
Movement of workers and consumers

Trucks, ships, aircraft and conventional vehicles commonly burn petroleum fuels and emit CO_2. Industrialization also encourages the construction of roads, ports, airports and logistics centres, increasing both energy use and land-surface modification.

The IPCC reported that transport accounted for approximately 15% of net global anthropogenic greenhouse-gas emissions in 2019.

Industrialization, urbanization and construction
Factories attract workers, services, housing and transport infrastructure. Industrialization therefore frequently causes rapid urbanization.

Industrialization and urban development may replace natural vegetation and soil with buildings, industrial estates, concrete structures, roofs, roads, bitumen surfaces and parking areas.

The urban heat-island effect mainly modifies local and regional temperature. The principal global climatic effect of industrialization results from greenhouse-gas emissions.

Deforestation and land-use change
Industrialization may cause forests to be cleared for mining, factories, roads and railways, dams and power projects, industrial townships, commercial agriculture and raw-material plantations.

Deforestation affects climate in two main ways:

1. carbon stored in vegetation may be released;
2. the removal of trees reduces future CO_2 absorption.

Land-cover change also modifies surface albedo, evapotranspiration, soil moisture, local temperature, and runoff and rainfall patterns.

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Increased material consumption

Industrialization promotes the mass production and consumption of vehicles, buildings, appliances, machinery, packaging materials, electronics, plastics and infrastructure.

Each product has an associated chain of emissions:

Raw-material extraction
↓
Material processing
↓
Manufacturing
↓
Packaging
↓
Transportation
↓
Use of the product
↓
Disposal or recycling

This complete sequence is known as a product life cycle. Climate emissions may occur at every stage. Increased demand for steel, cement, plastics and other manufactured materials has been an important driver of industrial emissions.

Direct and indirect emissions from industrialization

Emission category	Description	Example
Direct fuel emissions	Emissions produced by fuel burned within an industrial facility	Coal burned in a factory boiler
Process emissions	Emissions produced by chemical or physical transformation	CO_2 released during limestone calcination
Indirect energy emissions	Emissions from electricity or heat purchased by industry	Coal-based electricity used in a factory
Upstream emissions	Emissions from extracting and transporting fuels and raw materials	Diesel used in mining
Downstream emissions	Emissions from transporting, using or disposing of industrial products	Fuel used to transport manufactured goods
Land-use emissions	Emissions caused by clearing or modifying land	Forest removed for a mine or industrial estate

Major greenhouse gases linked with industrialization

Greenhouse gas	Industrial connection	Climate significance
Carbon dioxide, CO_2	Fossil-fuel burning, cement, steel, electricity and transport	Principal long-lived greenhouse gas associated with industrialization
Methane, CH_4	Coal mining and fossil-fuel extraction, processing and transport	Strong heat-trapping gas
Nitrous oxide, N_2O	Some chemical and industrial processes and associated agricultural expansion	Long-lived greenhouse gas
HFCs	Industrial and technological product use	Strong heat-trapping fluorinated gases
PFCs	Certain industrial processes	Long atmospheric persistence
SF_6	Specialised electrical and industrial equipment	Very strong heat-trapping gas

How industrial emissions strengthen the greenhouse effect

The natural greenhouse effect maintains Earth at a habitable temperature. Industrialization strengthens this natural process by increasing greenhouse-gas concentrations.

Solar shortwave radiation enters the atmosphere
↓
Earth's surface absorbs energy and becomes warm
↓
Earth emits longwave infrared radiation
↓
Industrial greenhouse gases absorb more infrared energy
↓
Part of the energy is re-emitted towards Earth
↓
Less heat escapes directly to space
↓
Surface, atmosphere and oceans become warmer

Atmospheric evidence of industrial-era influence

Gas	Pre-industrial concentration	Later concentration reported in the lecture	Approximate increase
CO_2	280 ppm	414 ppm in 2020	48%
CH_4	722 ppb	1,867 ppb in 2019	159%
N_2O	270 ppb	332 ppb in 2019	23%

These values show a rapid rise in greenhouse-gas concentrations following industrial development.

The IPCC states that the observed increases in well-mixed greenhouse gases since around 1750 are unequivocally caused by human activities.

Industrialization and global emissions

Industrialization affects several economic sectors simultaneously.

According to the IPCC's 2019 sectoral assessment:

Sector	Approximate share of net global GHG emissions
Energy supply	34%
Industry	24%
Agriculture, forestry and other land use	22%
Transport	15%
Buildings	6%

The percentages should not simply be added to estimate "industrialization emissions," because industrialization influences all these sectors and electricity-related emissions may be allocated differently depending on the accounting approach.

Why industrialization produces persistent climate change

Industrialization does not cause only temporary local pollution; it influences the climate over long periods. Greenhouse gases accumulate in the atmosphere, and some remain there for decades or centuries. Oceans absorb and store large quantities of heat, while industrial infrastructure may continue operating and producing emissions for several decades. Expanding production increases cumulative emissions, and deforestation weakens natural carbon sinks.

Warming may also activate feedback mechanisms, such as an increase in atmospheric water vapour, which can further intensify climate change. Therefore, the climate effect depends strongly on cumulative emissions, not only on emissions from one year.

Industrialization and climate feedbacks

Water-vapour feedback

Industrial GHG emissions
↓
Initial warming
↓
More evaporation
↓
More atmospheric water vapour
↓
Additional greenhouse warming

A warmer atmosphere can hold approximately 7% more moisture per 1°C of warming. Water vapour then amplifies the original warming caused by anthropogenic greenhouse gases.

Snow–ice albedo feedback

Industrial emissions
↓
Global warming
↓
Snow and ice melt
↓
Darker surfaces are exposed
↓
More solar radiation is absorbed
↓
Further warming

Carbon-sink feedback

Warming can influence the capacity of oceans and ecosystems to absorb carbon. The ocean as an important CO_2 sink and notes that temperature change may alter natural greenhouse-gas sources and sinks.

Consequences of industrialization-induced warming

Industrial emissions contribute to global warming, leading to higher atmospheric, land and ocean temperatures, as well as more frequent and intense heat waves. The resulting climate changes cause melting of glaciers and sea ice, sea-level rise, altered rainfall patterns, heavier precipitation, drought and increased wildfire risk. They also affect water resources, damage agriculture and ecosystems, and create serious risks to human health and infrastructure.

Human-induced warming raises the background temperature on which natural weather systems operate. Consequently, naturally occurring heat waves, storms or droughts may become more intense or cross earlier environmental and engineering thresholds.

Fossil-Fuel Use and Deforestation

Fossil-fuel use and deforestation are two major human causes of climate change. Fossil-fuel use releases carbon that has remained stored beneath Earth's surface for millions of years, whereas deforestation releases carbon stored in trees, vegetation and soils. Deforestation also reduces the ability of forests to remove carbon dioxide from the atmosphere. Both activities increase atmospheric greenhouse-gas concentrations, disturb the natural carbon cycle and alter Earth's energy balance. Industrialisation, urbanisation, deforestation and land-use change are therefore closely connected causes of anthropogenic climate change.

Fossil-Fuel Use
Fossil fuels are carbon-rich energy resources formed from the remains of ancient plants and organisms that were buried and transformed by heat and pressure over geological time. Coal, petroleum and natural gas are the three principal fossil fuels. They contain concentrated chemical energy that is released through combustion. Fossil fuels are widely used for electricity generation, industrial production, transportation, heating, construction and commercial activities.

Major Uses of Fossil Fuels
Electricity and Heat Production
Coal, petroleum and natural gas are extensively burned in thermal power plants to generate electricity. They are also used in boilers and heating systems to produce steam and heat for residential, commercial and industrial purposes. In coal-based and gas-based power plants, the heat released during combustion is used to convert water into steam, which drives turbines connected to electrical generators.

Transportation
The transportation sector depends heavily on petroleum-based fuels. Petrol and diesel are used in cars, motorcycles, buses, trucks and construction machinery. Diesel is also used in railway locomotives, while ships commonly use fuel oil or diesel. Aircraft depend mainly on aviation turbine fuel. The combustion of these fuels releases carbon dioxide and other pollutants directly into the atmosphere.

Industrial Activities
Industries use fossil fuels to operate machinery, heat furnaces and boilers, produce steam, melt materials, dry products and generate electricity. Coal and natural gas are particularly important in cement, iron and steel, chemical, fertiliser, glass, ceramic and petrochemical industries. Industrial fossil-fuel use therefore produces both direct combustion emissions and indirect emissions through purchased electricity.

Buildings
Coal, oil and natural gas may also be used in residential and commercial buildings for cooking, water heating, space heating and electricity generation. In areas where electricity is generated from fossil fuels, the use of electrical appliances, air-conditioning and heating systems also indirectly contributes to greenhouse-gas emissions.

Fossil-Fuel Combustion
Combustion is a chemical reaction in which a fuel reacts with oxygen and releases heat energy. Carbon present in a fossil fuel combines with oxygen to produce carbon dioxide. The simplified combustion of carbon can be represented as:

$$C + O_2 \rightarrow CO_2 + \text{energy}$$

Natural gas consists mainly of methane. Its complete combustion can be represented as:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{energy}$$

Although coal, petroleum and natural gas have different chemical compositions, all three produce carbon dioxide when burned. Fossil-fuel combustion is therefore a major human source of atmospheric carbon dioxide.

How Fossil-Fuel Use Disturbs the Carbon Cycle

Under the natural carbon cycle, carbon continuously moves among the atmosphere, oceans, vegetation, soils and rocks. Fossil fuels contain carbon that has remained isolated beneath Earth's surface for millions of years. When coal, petroleum and natural gas are extracted and burned, this geological carbon is rapidly transferred into the atmosphere as carbon dioxide.

Oceans, forests and soils absorb part of the additional carbon dioxide, but natural carbon sinks cannot remove it as rapidly as human activities release it. The remaining carbon dioxide accumulates in the atmosphere and strengthens the greenhouse effect. Fossil-fuel combustion therefore introduces additional carbon into the active carbon cycle rather than merely recycling carbon already present in vegetation and the atmosphere.

Greenhouse Gases Associated with Fossil-Fuel Use

Carbon Dioxide

Carbon dioxide is the principal greenhouse gas released during fossil-fuel combustion. Major sources include coal-fired thermal power plants, petrol and diesel vehicles, industrial furnaces, boilers, generators and fossil-fuel-based heating systems. The industrial-era increase in atmospheric carbon dioxide to fossil-fuel burning and notes an increase from approximately 280 parts per million in the eighteenth century to about 414 parts per million in 2020.

Methane

Methane is released during the extraction, processing and distribution of fossil fuels. Underground coal mining can release methane trapped within coal seams. Oil and natural-gas operations may release methane during drilling, processing, storage and transportation. Leakage may also occur from pipelines, compressors, valves, abandoned wells and gas-distribution systems. Venting and incomplete flaring further contribute to methane emissions. These emissions may occur before the fuel reaches its final consumer.

Nitrous Oxide

Small quantities of nitrous oxide may be generated during high-temperature fossil-fuel combustion. Power plants, industrial facilities and motor vehicles can therefore release nitrous oxide in addition to carbon dioxide and other combustion products.

Black Carbon and Other Pollutants

Incomplete combustion of coal, diesel, fuel oil and biomass can produce black carbon. Black carbon absorbs solar radiation and contributes to atmospheric warming. When deposited on snow and ice, it darkens the surface, reduces reflectivity and accelerates melting. Fossil-fuel combustion also produces sulphur dioxide, nitrogen oxides, particulate matter and other pollutants. Some atmospheric particles temporarily reflect sunlight and cause cooling, but this short-lived effect does not cancel the long-term warming caused by accumulated greenhouse gases.

Life-Cycle Emissions from Fossil Fuels
Exploration
Energy is consumed while locating and assessing coal, petroleum and natural-gas reserves. Geological surveys, exploratory drilling, transportation and supporting infrastructure all require fuel and electricity.

Extraction
Coal mining, oil drilling and natural-gas extraction require heavy machinery, pumps, drilling equipment and ventilation systems. These activities consume fossil fuels and electricity. Coal mines and oil-and-gas operations may also directly release methane into the atmosphere.

Processing
Fossil fuels often require processing before they can be used. Crude oil is refined into petrol, diesel, kerosene and other petroleum products. Natural gas is processed to remove impurities, while coal may be washed, crushed, graded and prepared for combustion. These processes require additional energy and produce greenhouse-gas emissions.

Transportation and Distribution
Coal, oil and natural gas are transported through pipelines, railway wagons, trucks, ships and oil tankers. The transportation system consumes energy and generates emissions. Leakage and accidental releases may also occur during the storage and distribution of oil and natural gas.

Final Combustion
The greatest quantity of carbon dioxide is generally released when fossil fuels are finally burned in power plants, vehicles, industries, generators and buildings. The complete climate impact of fossil fuels therefore includes emissions from exploration, extraction, processing, transportation, distribution and final use.

Fossil Fuels and the Enhanced Greenhouse Effect
Solar radiation reaches Earth mainly in the form of short-wave radiation. Some of this energy is reflected by the atmosphere, clouds and surface, while the remainder is absorbed and warms the planet. The heated Earth emits energy back towards space as long-wave infrared radiation. Greenhouse gases absorb and re-emit part of this outgoing infrared radiation. When fossil-fuel use increases the concentration of carbon dioxide and other greenhouse gases, a greater amount of outgoing energy is retained within the lower atmosphere and surface system. This strengthens the natural greenhouse effect and raises global temperature.

The process can be represented as:
Fossil-fuel burning → increased greenhouse-gas concentrations → greater absorption of outgoing infrared radiation → additional heat retained in the Earth system → global warming

The greenhouse gases behave like a blanket by allowing much of the incoming short-wave radiation to pass through while absorbing part of the outgoing long-wave radiation.

Cumulative Nature of Fossil-Fuel Emissions
The climatic effect of carbon dioxide depends strongly on cumulative emissions. Each additional quantity of coal, oil or natural gas burned adds more carbon dioxide to the

atmosphere. Therefore, continued emissions produce further warming even when annual emissions remain approximately constant.

Power plants, factories, transport networks, buildings and fossil-fuel supply systems often operate for several decades. Continued investment in such systems can create carbon lock-in, meaning that societies remain dependent on fossil fuels because large investments have already been made in carbon-intensive infrastructure. Reducing future warming therefore requires both limiting new fossil-fuel infrastructure and replacing existing systems with low-carbon alternatives.

Climate Effects of Fossil-Fuel Use

Fossil-fuel emissions contribute to increasing atmospheric, land and ocean temperatures. Higher temperatures increase the frequency and intensity of heat waves and contribute to the melting of glaciers, ice sheets and sea ice. Ocean warming causes thermal expansion of seawater, while melting land ice further contributes to sea-level rise.

Fossil-fuel-driven warming also alters rainfall patterns and increases the capacity of the atmosphere to hold moisture. This may intensify heavy precipitation in some regions while higher temperatures and increased evaporation contribute to drought in others. Hotter and drier conditions increase wildfire risk. Additional impacts include ocean acidification, ecosystem disruption, reduced agricultural productivity, water scarcity, health risks and damage to buildings and infrastructure.

Deforestation

Deforestation refers to the permanent or long-term removal of forest vegetation followed by the conversion of forest land to another use. Forests may be cleared for commercial agriculture, cattle grazing, settlements, industries, mining, roads, railways, dams, power projects, timber extraction and urban development.

Deforestation differs from temporary tree harvesting where the forest is allowed to regenerate. In deforestation, the original forest is generally replaced by cropland, pasture, buildings, roads, mines or another form of land use.

Forests as Carbon Reservoirs and Carbon Sinks

Forests store large quantities of carbon in trunks, branches, leaves, roots, dead organic matter, forest litter and soils. Through photosynthesis, trees absorb carbon dioxide from the atmosphere and convert it into organic matter.

The process of photosynthesis can be represented as:

$$6CO_2 + 6H_2O + \text{sunlight} \rightarrow C_6H_{12}O_6 + 6O_2$$

A growing forest acts as a carbon sink when it absorbs more carbon dioxide than it releases. Forests influence climate in two major ways. First, they store carbon that has already been removed from the atmosphere. Second, they continue to absorb additional carbon dioxide as trees and other vegetation grow.

How Deforestation Releases Carbon Dioxide

Burning of Vegetation

Forests are often burned after clearing to remove unwanted vegetation and prepare land for agriculture or development. Burning rapidly converts carbon stored in trees, leaves and other vegetation into carbon dioxide. Incomplete burning may also release methane and black carbon.

Decomposition of Cleared Biomass

When felled trees, roots, branches and leaves are left on the ground, microorganisms gradually decompose the organic matter. Carbon stored in the biomass is then released mainly as carbon dioxide.

Use of Wood Products

Some carbon may remain temporarily stored in timber, furniture and other wood products. However, this carbon is eventually released when the products decay, are discarded or are burned.

Soil Disturbance

Forest soils contain substantial quantities of organic carbon. Clearing, ploughing, drainage and exposure of the soil increase aeration and accelerate the decomposition of soil organic matter. As decomposition increases, additional carbon dioxide is released into the atmosphere.

Loss of Carbon-Sequestration Capacity

The climatic effect of deforestation is not limited to the immediate release of stored carbon. After trees are removed, the land loses much of its capacity to absorb carbon dioxide through photosynthesis. Deforestation therefore has a double effect: it releases carbon that was previously stored in vegetation and soils, and it reduces future removal of carbon dioxide from the atmosphere.

Forest degradation has a similar effect even when the forest is not completely removed. Logging, repeated fires, livestock grazing, fragmentation and other disturbances reduce forest biomass and weaken its carbon-storage capacity.

Effect of Deforestation on Surface Albedo

Albedo is the proportion of incoming solar radiation reflected by a surface. Forests are generally darker than many grasslands, croplands and exposed soils. Removing forests changes the reflectivity of the land surface and consequently alters the amount of solar energy absorbed. The direction and magnitude of this effect depend on latitude, forest type, season, snow cover and the type of land cover that replaces the forest. In some snow-covered high-latitude regions, removing dark trees may expose reflective snow and increase local albedo. However, this possible regional cooling effect does not remove the warming caused by carbon emissions and the loss of forest carbon storage.

Effect on Evapotranspiration and Temperature

Trees absorb water through their roots and release water vapour through their leaves. The combined transfer of water from soil and vegetation to the atmosphere is known as evapotranspiration. This process uses heat energy and provides natural cooling.

When forests are cleared, evapotranspiration and evaporative cooling decline. A greater proportion of available solar energy then heats the land and surrounding air. As a result, surface temperatures may rise, soil moisture may decrease and local temperature variations may become more extreme. The deforestation and other land-surface changes modify the reflection and absorption of solar energy and may increase local and regional temperatures.

Effect on Rainfall and the Water Cycle

Forests play an important role in the water cycle. Water vapour released through evapotranspiration contributes to atmospheric moisture, cloud formation and rainfall recycling. Large forested regions can influence rainfall both within the forest and in areas located downwind.

Deforestation reduces evapotranspiration and may lower atmospheric moisture and regional rainfall. It also reduces the interception of rainfall by leaves and branches. Without forest vegetation, more rainwater flows rapidly over the land surface instead of infiltrating into the soil. This may increase surface runoff, erosion, sediment transport and flood peaks while reducing groundwater recharge and dry-season streamflow.

Forest Fires and Climate Feedback

Deforestation and forest degradation may make the remaining forest hotter, drier and more vulnerable to fire. Forest fires release carbon dioxide, methane, black carbon and other pollutants into the atmosphere. Repeated fires also reduce vegetation cover and weaken the ability of forests to recover.

A reinforcing feedback can therefore develop:
Deforestation → reduced moisture and higher temperature → increased fire risk → further forest loss → additional greenhouse-gas emissions
Global warming may intensify this cycle by increasing heat stress, drought, pest outbreaks and tree mortality. As forests weaken, their ability to absorb and store carbon also declines.

Forest Fragmentation

Forest fragmentation occurs when a large continuous forest is divided into smaller patches by roads, farms, settlements, mines, transmission lines or other development. Fragmented forests have a greater proportion of exposed edges.

Conditions at forest edges are usually hotter, drier and windier than conditions inside an undisturbed forest. These conditions may increase tree mortality, reduce biomass, promote fire penetration and weaken carbon storage. Roads through forests also create access for further logging, settlement and land clearing. Thus, a fragmented forest may store less carbon even when it has not been completely removed.

Relationship Between Fossil-Fuel Use and Deforestation

Fossil-fuel use and deforestation are closely interconnected. Fossil fuels provide the energy required for cutting trees, constructing roads, operating mining machinery, transporting timber, processing agricultural products and developing industries and settlements on cleared land.

At the same time, forests may be removed to develop coal mines, oil and gas fields, pipelines, power plants, dams, industrial estates and transportation networks. Fossil-fuel use introduces geological carbon into the active carbon cycle, whereas deforestation releases biological carbon and weakens natural carbon sinks.

Fossil-fuel emissions arise mainly through combustion, although methane may also be released during extraction and distribution. Deforestation emissions arise through burning, decomposition and soil disturbance. Deforestation additionally modifies albedo, evapotranspiration, rainfall, runoff and local temperature.

Combined Causal Chain

Fossil-Fuel Pathway

The fossil-fuel pathway begins with the exploration and extraction of coal, petroleum and natural gas. The fuels are then processed, transported and distributed to power plants, industries, vehicles and buildings. Their combustion releases carbon dioxide, while extraction and distribution may release methane. These greenhouse gases strengthen the greenhouse effect, cause additional heat to accumulate in the Earth system and drive global warming and climate change.

Deforestation Pathway

The deforestation pathway begins with forest clearing and degradation. Burning and decomposition release carbon stored in vegetation, while soil disturbance releases additional carbon. The removal of trees reduces future carbon-dioxide absorption and changes albedo, evapotranspiration, temperature and rainfall. These changes contribute to both global warming and regional climate modification.

Measures for Reducing These Causes

Reducing Fossil-Fuel Emissions

Fossil-fuel emissions can be reduced by increasing the use of solar, wind and other low-carbon energy sources. Energy efficiency in buildings, industries, appliances and transportation can reduce the total quantity of fuel required. Electrification of vehicles and industrial processes can reduce direct fossil-fuel combustion when the electricity is generated from low-carbon sources.

Other important measures include strengthening public transport, reducing unnecessary energy consumption, improving industrial processes, controlling methane leakage, eliminating routine venting and flaring, and gradually replacing coal-, oil- and gas-based infrastructure with low-carbon alternatives. The reduced fossil-fuel burning and a shift towards renewable energy as important mitigation measures.

Reducing Deforestation

Deforestation can be reduced through the protection of natural forests, control of illegal logging and improvement of forest-fire management. Sustainable forest management can allow the use of forest resources while maintaining forest cover, biodiversity and carbon storage.

Restoration of degraded forests, reforestation and afforestation can increase carbon sequestration. Agricultural productivity should be improved on existing farmland so that additional forest land is not cleared. Reducing wasteful consumption of timber and paper, recognising the rights of forest-dependent communities and improving satellite monitoring of land-use change can also support forest conservation.

Forest protection is an essential component of climate mitigation, but it cannot substitute for reductions in fossil-fuel emissions. Both energy-system transformation and forest conservation are necessary.

Global Warming

Global warming refers to the long-term increase in the average temperature of Earth's surface and lower atmosphere. It is one component of climate change. Climate change is a broader term that includes changes in temperature, rainfall, winds, oceans, glaciers, sea ice, sea level and the frequency or intensity of extreme events.

A global warming trend is the long-term direction and rate of change in global temperature. It is determined using observations collected over decades rather than the temperature of a single day, month or year.

Global warming does not mean that every place becomes warmer every day. Regional temperatures can temporarily fall because of weather systems, ocean circulation, volcanic eruptions or natural climate variability. However, these short-term changes occur around a rising long-term global temperature trend.

Temperature Anomaly

Global warming is generally expressed using a temperature anomaly rather than an absolute temperature. A temperature anomaly is the difference between the temperature during a particular period and the average temperature during a selected reference period.

Temperature anomaly=Observed temperature−Reference-period temperature

For example, a temperature anomaly of +1.5°C means that the measured global temperature was 1.5°C higher than the average temperature during the reference period. The IPCC and WMO commonly use 1850–1900 to represent pre-industrial climatic conditions. This period is used because it predates most of the rapid increase in industrial greenhouse-gas emissions.

How Global Temperature Is Measured

Global mean near-surface temperature combines:

- air temperature measurements over land;
- sea-surface temperature measurements over oceans; and
- additional observations incorporated through climate reanalysis systems.

Land air temperatures are commonly measured at weather stations approximately 1.5–2 metres above the ground. Sea-surface temperatures are measured using ships, buoys and other ocean-observing systems. The observations are quality-controlled, adjusted for changes in instruments and measurement practices, and combined statistically to produce a global mean.

The WMO assessment for 2024 used six major global temperature datasets: Berkeley Earth, ERA5, GISTEMP, HadCRUT5, JRA-3Q and NOAAGlobalTemp. Although there are small differences among the datasets, they show substantially the same long-term warming pattern.

Historical Global Temperature Trend

Instrumental records show fluctuations from year to year, but the long-term trend is clearly upward. The IPCC reports that the average global surface temperature during 2011–2020 was approximately 1.09°C above the 1850–1900 average. The first two decades of the twenty-first century, 2001–2020, were approximately 0.99°C warmer than 1850–1900.

The warming trend has accelerated. Global surface temperature has increased faster since 1970 than during any other 50-year period in at least the previous 2,000 years.

General pattern of the instrumental record

The long-term temperature record can be divided broadly into four stages:

Period	General temperature behaviour
1850–1910	Fluctuating temperatures with limited long-term increase
1910–1940	Noticeable warming
1940–1970	Slower warming and considerable variability
Since 1970	Rapid and sustained warming
Since 2015	Concentration of the warmest years in the observational record

The WMO temperature graph from 1850 to 2024 shows annual fluctuations around a strongly rising long-term curve, with the steepest sustained increase occurring during recent decades.

Recent Record Warming

The annually averaged global mean near-surface temperature in 2024 was 1.55°C ± 0.13°C above the 1850–1900 average. It was the warmest year in the 175-year observational record. The previous record was established in 2023, when global temperature was 1.45°C ± 0.12°C above the pre-industrial average. Every year from 2015 to 2024 ranked among the ten warmest years in the observational record.

Table 1. Selected global temperature indicators

Indicator	Temperature change above 1850–1900
Average for 2001–2020	0.99°C
Average for 2011–2020	1.09°C
Annual average for 2023	1.45°C ± 0.12°C
Annual average for 2024	1.55°C ± 0.13°C

The 2001–2020 and 2011–2020 values are multi-year averages, whereas the 2023 and 2024 values are individual annual averages. They should not be interpreted as directly equivalent climate periods.

Warming Over Land and Oceans

Warming is not geographically uniform. During 2011–2020:

- global surface warming was approximately 1.09°C;
- warming over land was approximately 1.59°C; and
- warming over the ocean surface was approximately 0.88°C.

Thus, the land surface has warmed more rapidly than the ocean surface.

The difference between land and ocean warming does not mean that the oceans are unaffected. The oceans absorb and store very large quantities of heat. The IPCC estimates that ocean warming accounted for approximately 91% of the heating of the climate system, while land warming accounted for about 5%, ice loss about 3%, and atmospheric warming about 1%.

Human Causes of the Observed Warming Trend

The IPCC concludes that human activities, principally greenhouse-gas emissions, have unequivocally caused global warming. The best estimate is that essentially all observed warming between 1850–1900 and 2010–2019 was caused by human influence. The likely range of human-caused global surface warming up to 2010–2019 was 0.8–1.3°C, with a best estimate of 1.07°C.

The contributions were assessed as follows:

Climate driver	Assessed temperature contribution
Well-mixed greenhouse gases	Warming of 1.0–2.0°C
Other human drivers, mainly aerosols	Cooling of 0.0–0.8°C
Natural solar and volcanic drivers	Approximately ±0.1°C
Internal climate variability	Approximately ±0.2°C
Best estimate of total human-caused warming	1.07°C

Greenhouse-gas warming has therefore been partly masked by the cooling effect of some human-produced aerosols. Natural solar and volcanic changes cannot explain the magnitude of the observed long-term warming.

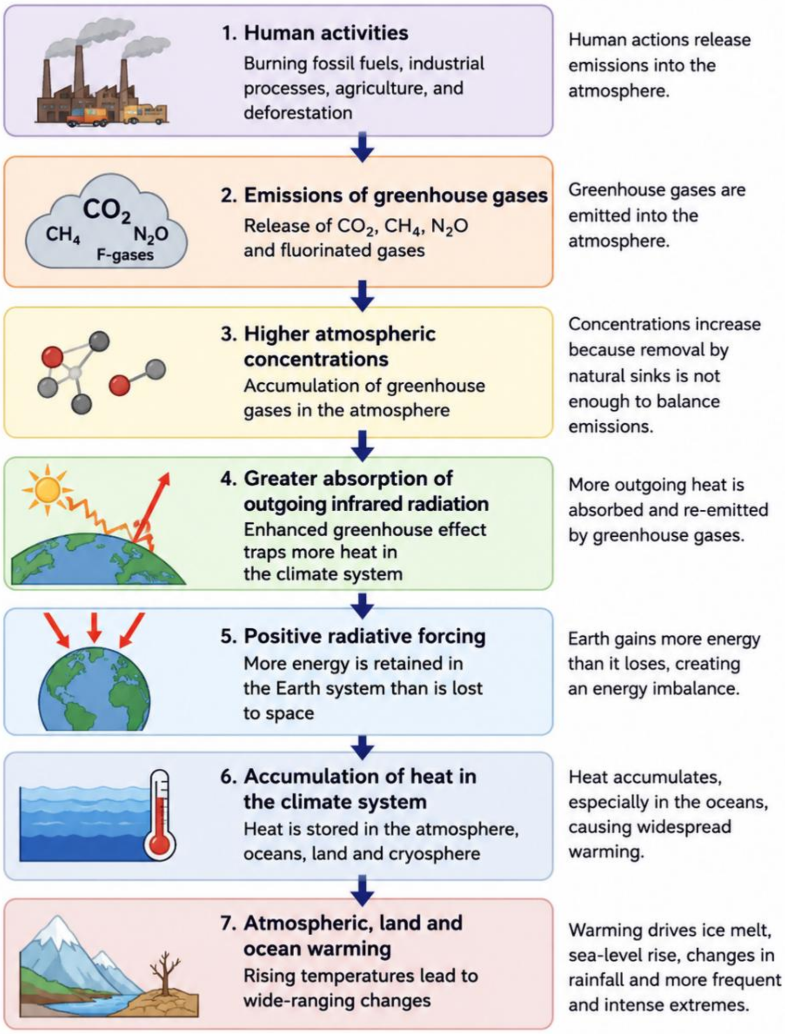

Figure 1. Causal chain of global warming

Greenhouse-Gas Concentration Trends

The warming trend is closely associated with the rapid increase in atmospheric greenhouse-gas concentrations since industrialisation.

In 2019, atmospheric concentrations reached approximately:

- CO_2: 410 ppm;
- CH_4: 1,866 ppb; and
- N_2O: 332 ppb.

The IPCC reports that concentrations of methane and nitrous oxide are higher than at any time in at least 800,000 years, while current carbon-dioxide concentrations are higher than at any time in at least two million years.

By 2023:

- CO_2 had reached 420.0 ± 0.1 ppm;
- CH_4 had reached $1,934 \pm 2$ ppb; and
- N_2O had reached 336.9 ± 0.1 ppb.

These values represented approximately 151%, 265% and 125% of their respective pre-industrial concentrations. Real-time observations indicated that all three gases continued to increase during 2024.

Table 2. Greenhouse-gas concentration trend

Gas	IPCC value for 2019	WMO value for 2023	Climate significance
Carbon dioxide	410 ppm	420.0 ppm	Largest long-term anthropogenic climate driver
Methane	1,866 ppb	1,934 ppb	Powerful greenhouse gas with strong near-term warming
Nitrous oxide	332 ppb	336.9 ppb	Long-lived greenhouse gas contributing persistent warming

The WMO identifies fossil-fuel burning, cement production and land-use change, including deforestation, as major anthropogenic sources of carbon dioxide. Vegetation and the oceans act as carbon sinks. During 2014–2023, an estimated 48% of anthropogenic carbon-dioxide emissions remained in the atmosphere, while the ocean and land sinks absorbed substantial shares.

Long-Term Trend and Short-Term Variability
Global temperature does not rise by exactly the same amount every year. The long-term human-caused warming trend is modified temporarily by natural variability.

Important natural influences include:

- El Niño and La Niña;
- volcanic eruptions;
- changes in ocean circulation; and
- short-term solar variability.

El Niño usually produces a temporary increase in global average temperature, while La Niña generally produces temporary cooling. The largest effect often appears a few months after the El Niño or La Niña peak because heat must move from the ocean into the atmosphere.

The strong 2023–2024 El Niño contributed to the record temperatures of 2023 and 2024. However, the El Niño occurred on top of the long-term warming caused by increasing greenhouse-gas concentrations. It did not create the underlying multi-decadal trend.

Key distinction
Long-term warming trend = mainly caused by increasing greenhouse gases

Year-to-year fluctuation around the trend = influenced by El Niño, La Niña, volcanoes and ocean circulation

Does One Year Above 1.5°C Mean That the Paris Limit Has Been Crossed?
No. The 2024 annual temperature was approximately 1.55°C above the 1850–1900 average, but a single year above 1.5°C does not mean that the long-term Paris Agreement warming level has formally been exceeded.

Climate change and global warming levels are evaluated over extended periods. The IPCC defines future global warming levels using 20-year averages relative to 1850–1900. A long-term exceedance of 1.5°C would therefore require the average temperature over a sustained multi-year period to reach that level.

This distinction can be written as:
Annual temperature anomaly > 1.5°C $\neq$ Confirmed long-term global warming of 1.5°C
An individual year can be temporarily raised by El Niño or reduced by La Niña or volcanic aerosols. The long-term warming level is the underlying climatic trend after short-term variability is considered.

Ocean Heat-Content Trend
Surface temperature is only one measure of global warming. Ocean heat content is a more direct indicator of the amount of excess energy accumulating in the climate system.
In 2024, global ocean heat content reached the highest level in the 65-year observational record. Each of the eight years leading up to 2024 established a new ocean heat-content record.

The rate of ocean warming during 2005–2024 was more than twice the rate observed during 1960–2005.

Period	Ocean warming rate
1960–2005	3.1–3.9 ZJ per year
2005–2024	11.2–12.1 ZJ per year

The rapid growth of ocean heat content confirms that global warming is not limited to atmospheric temperature measurements.

Consequences of increasing ocean heat
Increasing ocean heat contributes to:
- thermal expansion of seawater;
- rising sea level;
- marine heatwaves;
- coral-reef stress;
- changes in ocean circulation;
- reduced oxygen availability; and
- stronger impacts from coastal and marine extremes.

Global Sea-Level Trend
Global mean sea level has risen because ocean water expands as it warms and because glaciers and ice sheets lose mass.

The IPCC reports that global mean sea level increased by approximately 0.20 metres between 1901 and 2018. The rate of rise accelerated over time:

Period	Average sea-level-rise rate
1901–1971	1.3 mm/year
1971–2006	1.9 mm/year
2006–2018	3.7 mm/year

Human influence was very likely the main driver of this acceleration from at least 1971 onward.

Satellite observations show further acceleration. The WMO reports that the rate increased from 2.1 mm/year during 1993–2002 to 4.7 mm/year during 2015–2024. Global mean sea level reached a satellite-era record in 2024.

This means that the recent rate of global sea-level rise is more than twice the rate observed at the beginning of the satellite record.

Glacier-Mass Trend
Glaciers gain mass through snowfall and lose mass through melting, sublimation and calving. Glacier mass balance represents the difference between accumulation and loss.

The period from 2021/2022 to 2023/2024 produced the most negative three-year glacier mass balance in the observational record. Seven of the ten most negative annual glacier mass-balance years since 1950 have occurred since 2016.

Preliminary observations indicated that the average 2023/2024 mass balance of the global reference glaciers was close to −1.1 metres water equivalent. Only two of 141 reporting glaciers showed a positive mass balance at the time of assessment.

One metre water equivalent represents approximately one tonne of water per square metre of glacier area.

Glacier melting contributed approximately 21% of observed sea-level rise during 1993–2018. The IPCC assesses human influence as very likely the main driver of global glacier retreat since the 1990s.

Sea-Ice Trends
Arctic sea ice
The minimum Arctic sea-ice extent in 2024 was 4.28 million km^2, recorded on 11 September. It was the seventh-lowest minimum in the 46-year satellite record and was 1.17 million km^2 below the 1991–2020 average minimum. All 18 of the lowest Arctic sea-ice minima in the satellite record occurred during the most recent 18 years.

The long-term decline in the minimum Arctic sea-ice extent from 1979 to 2024 was approximately 14% of the 1991–2020 average per decade, equivalent to a loss of around 77,000 km^2 per year.

Antarctic sea ice

The minimum Antarctic sea-ice extent in 2024 was 1.99 million km², tied for the second-lowest minimum in the satellite record. It was the third consecutive year in which the Antarctic minimum fell below 2 million km².

The 2024 Antarctic maximum extent was approximately 17.16 million km², also the second lowest on record.

Ice–albedo feedback

Warming
↓
Snow and ice melt
↓
Darker ocean or land surface is exposed
↓
Less solar energy is reflected
↓
More solar energy is absorbed
↓
Additional warming and melting

This is a positive feedback because the initial warming produces changes that further increase warming.

Ocean-Acidification Trend

Ocean acidification is not a temperature trend, but it is an important indicator of increasing atmospheric carbon dioxide.

Approximately one quarter of the carbon dioxide emitted by human activities during 2014–2023 was absorbed by the oceans. This altered seawater carbonate chemistry and reduced surface-ocean pH.

Global average surface-ocean pH decreased at approximately 0.017 ± 0.001 pH units per decade during 1985–2023. Acidification was not uniform, with some regions of the Indian, Southern, Pacific and Atlantic oceans experiencing faster declines than the global average.

Ocean acidification provides additional evidence that anthropogenic carbon dioxide is changing the entire Earth system, not merely atmospheric temperature.

Regional Temperature Patterns

Global warming is a worldwide phenomenon, but its magnitude and expression vary by region.

In 2024, most land areas were warmer than the 1991–2020 average. Record or near-record annual temperatures were recorded across large tropical regions extending from South and Central America towards the western Pacific.

Exceptionally high annual temperatures were also observed in:

- eastern North America;
- North Africa;
- Europe;
- southern Asia; and
- eastern Asia.

Record sea-surface temperatures occurred in the tropical and North Atlantic, the tropical Indian Ocean, parts of the western Pacific and parts of the Southern Ocean.

Rainfall and Hydrological Changes

Global warming alters evaporation, atmospheric moisture, circulation and precipitation. However, rainfall changes are not uniform: some regions become wetter, while others experience declining rainfall and greater drought risk.

During 2024, drier-than-average conditions occurred across much of Southern Africa, parts of North and West Africa, large areas of South America, north-western Mexico, parts of northern North America, southern Australia and parts of southern Europe.

Wetter-than-average conditions occurred in parts of the Sahel, Central and East Africa, Central and Western Europe, northern and eastern Australia, and extensive areas of Asia.

One year's precipitation map should not by itself be treated as a long-term climate trend. It represents the combined effects of long-term climate change, El Niño, atmospheric circulation and regional weather variability.

Trends in Climate Extremes

Human-caused climate change is already affecting weather and climate extremes in every region. Evidence has strengthened for changes in:

- heatwaves;
- heavy precipitation;
- drought;
- fire weather;
- tropical-cyclone rainfall; and
- compound extreme events.

The IPCC reports that climate change is affecting extremes globally and producing widespread impacts on food security, water security, human health, ecosystems, economies and infrastructure.

Heat extremes

Hot extremes and heatwaves have become more intense in many regions. Numerous prolonged heatwaves occurred during 2024, affecting East Asia, south-eastern Europe, the Mediterranean, the Middle East, the south-western United States, South-east Asia, West Africa, Central America and northern India.

Heavy rainfall and floods

A warmer atmosphere can hold more water vapour, creating conditions for heavier rainfall when storms occur. Severe floods and extreme rainfall affected several regions during 2024, including East Africa, the Sahel, southern Brazil, Spain and parts of the United States.

Drought and wildfire

Higher temperatures increase evaporation and may intensify drying where rainfall is limited. Significant drought occurred in Southern Africa and parts of the Americas during 2024. Major wildfire activity affected Chile, Canada, the western United States and the Brazilian Amazon.

Future Global Warming Trend

Global warming is expected to continue increasing in the near term under nearly all assessed emission scenarios because of continuing and cumulative carbon-dioxide emissions. Natural variability may temporarily accelerate or slow the observed temperature increase in individual years, especially at regional scales. However, it has little effect on the long-term centennial warming trend.

The IPCC assesses that:

- the best estimate for reaching 1.5°C of long-term global warming lies in the first half of the 2030s under most assessed scenarios;
- warming of 2°C is likely to be exceeded during the twenty-first century unless deep reductions in carbon dioxide and other greenhouse gases occur;
- deep, rapid and sustained emission reductions are required to limit warming; and
- changes in the global surface-temperature trend following strong emission reductions may become discernible after approximately 20 years.

Consolidated Table of Global Warming Indicators

Indicator	Observed trend reported in the sources
Global surface temperature	Approximately 1.09°C above 1850–1900 during 2011–2020
2024 temperature	1.55°C ± 0.13°C above 1850–1900; warmest year on record
Recent warm years	2015–2024 were the ten warmest years on record
Land temperature	Approximately 1.59°C above 1850–1900 during 2011–2020
Ocean-surface temperature	Approximately 0.88°C above 1850–1900 during 2011–2020
Atmospheric CO_2	420.0 ppm in 2023; continued increase observed in 2024
Ocean heat content	Record high in 2024; eight consecutive annual records
Sea-level-rise rate	Increased from 2.1 mm/year in 1993–2002 to 4.7 mm/year in 2015–2024
Glacier mass balance	Most negative three-year balance on record during 2021/22–2023/24
Arctic sea ice	Declining approximately 14% per decade for minimum extent
Antarctic sea ice	2024 minimum and maximum were both second lowest on record
Ocean pH	Declining approximately 0.017 pH units per decade during 1985–2023
Climate extremes	Increasing evidence of intensified heat, heavy rainfall, drought and compound events

Climate Change Indicators

Climate change indicators are measurable physical, chemical and biological changes in the Earth system that provide scientific evidence that the climate is changing. Unlike individual weather events, these indicators represent long-term observations collected over many years or decades. They are monitored by organizations such as the Intergovernmental Panel on Climate Change (IPCC), the World Meteorological Organization (WMO), NASA and national meteorological agencies.

No single indicator alone proves climate change. Scientists evaluate multiple indicators together because they respond consistently to the warming caused by increasing greenhouse-gas concentrations. Rising global temperature, warming oceans, melting glaciers, shrinking sea ice, rising sea level and increasing greenhouse-gas concentrations all provide independent evidence of a warming climate.

Major Climate Change Indicators

1. Rising Global Temperature

The increase in global average surface temperature is the most direct indicator of climate change. Global temperature is determined using measurements from thousands of land-based weather stations, ocean buoys, ships and satellite observations.

The IPCC reports that the global surface temperature during 2011–2020 was approximately 1.09°C higher than the 1850–1900 pre-industrial average. According to the WMO, 2024 was the warmest year in the instrumental record, with a global temperature approximately 1.55°C above the pre-industrial average. The concentration of the warmest years during the last decade clearly demonstrates a persistent long-term warming trend rather than short-term natural variability.

Why it is important

- Indicates accumulation of heat in the climate system.
- Influences rainfall, droughts, storms and ecosystems.
- Serves as the primary indicator used to evaluate global warming targets.

2. Rising Atmospheric Greenhouse-Gas Concentrations

The rapid increase in atmospheric greenhouse gases provides direct evidence of human influence on the climate.

Since the beginning of industrialisation:

Greenhouse Gas	Pre-industrial	Recent Value
Carbon dioxide (CO_2)	280 ppm	~420 ppm (2023)
Methane (CH_4)	722 ppb	~1934 ppb
Nitrous oxide (N_2O)	270 ppb	~337 ppb

These increasing concentrations strengthen the greenhouse effect by trapping more outgoing infrared radiation, causing additional warming of the atmosphere, land and oceans.

3. Ocean Heat Content

More than 90% of the excess heat generated by global warming is absorbed by the oceans. Therefore, ocean heat content is considered one of the most reliable indicators of climate change.

The WMO reports that global ocean heat content reached a record high in 2024, and the warming rate during 2005–2024 was more than twice that observed during 1960–2005. Continued ocean warming contributes to marine heatwaves, coral bleaching, reduced dissolved oxygen and changes in ocean circulation.

Importance

- Confirms continued accumulation of heat in the climate system.
- Drives sea-level rise through thermal expansion.
- Influences global weather and climate patterns.

4. Glacier Melting and Ice Loss

Glaciers around the world are losing mass because melting exceeds snowfall accumulation. Recent observations show that the period 2021–2024 recorded the most negative three-year glacier mass balance since observations began. The IPCC concludes that human influence is the primary cause of global glacier retreat since the 1990s.

Melting glaciers reduce freshwater storage, alter river flows and contribute significantly to rising sea level.

Effects

- Reduced freshwater availability.
- Increased sea-level rise.
- Changes in river discharge.
- Greater risk of glacial lake outburst floods.

5. Declining Arctic and Antarctic Sea Ice

Satellite observations show a continuous decline in Arctic sea-ice extent.
The minimum Arctic sea ice in 2024 was among the lowest recorded, and all of the 18 lowest Arctic sea-ice minima have occurred during the most recent eighteen years. Antarctic sea ice has also shown exceptionally low extents during recent years.

Loss of sea ice reduces Earth's reflectivity (albedo), allowing more solar energy to be absorbed, which further accelerates warming through the ice–albedo feedback mechanism.

6. Rising Sea Level

Global mean sea level is increasing because:

- ocean water expands as it warms (thermal expansion);
- glaciers melt;
- Greenland and Antarctic ice sheets lose mass.

The IPCC estimates that global mean sea level rose by approximately 0.20 m between 1901 and 2018, while recent satellite observations show an accelerated rate of approximately 4.7 mm per year during 2015–2024.

Sea-level rise increases the frequency of coastal flooding, shoreline erosion, saltwater intrusion into groundwater and storm-surge damage.

7. Ocean Acidification

The oceans absorb a significant proportion of anthropogenic carbon dioxide emitted into the atmosphere.

When carbon dioxide dissolves in seawater, it forms carbonic acid, lowering ocean pH. Global average surface-ocean pH has steadily declined during recent decades, indicating increasing ocean acidification.

Ocean acidification affects coral reefs, shell-forming organisms, fisheries and marine biodiversity.

8. Changes in Extreme Weather Events
Climate change is increasing the frequency and intensity of many weather extremes.
Scientific observations indicate increases in:

- Heatwaves
- Heavy rainfall events
- Floods
- Droughts
- Wildfires
- Tropical cyclone rainfall
- Compound extreme events

Although individual weather events still occur naturally, global warming raises the background temperature and atmospheric moisture, making many extreme events more severe than they were in the past.

Climate Change Indicators

Indicator	Evidence of Climate Change	Major Consequences
Rising global temperature	Higher global average temperature	Heatwaves, ecosystem stress
Greenhouse-gas concentrations	Increasing CO_2, CH_4 and N_2O	Enhanced greenhouse effect
Ocean heat content	Record ocean warming	Marine heatwaves, sea-level rise
Glacier melting	Negative glacier mass balance	Reduced freshwater, sea-level rise
Sea-ice decline	Shrinking Arctic and Antarctic sea ice	Lower albedo, faster warming
Rising sea level	Thermal expansion and ice-sheet melting	Coastal flooding, erosion
Ocean acidification	Declining ocean pH	Coral bleaching, marine ecosystem damage
Climate extremes	More frequent heatwaves, floods and droughts	Infrastructure damage, agriculture and health impacts